



The effectiveness of premedication for endotracheal intubation in mechanically ventilated neonates A systematic review

Vibhuti Shah, MD, MRCP (UK)^{a,*},
Arne Ohlsson, MD, MSc, FRCPC, FAAP^b

^a*Department of Paediatrics, Mount Sinai Hospital, Faculty of Medicine, University of Toronto,
600 University Avenue, Toronto, ON M5G 1X5, Canada*

^b*Evidence-Based Neonatal Care and Outcomes Research, Department of Paediatrics,
Mount Sinai Hospital, Faculty of Medicine, University of Toronto,
600 University Avenue, Toronto, ON M5G 1X5, Canada*

There is mounting scientific evidence that in fetuses and neonates, the neuro-anatomic and neuroendocrine systems are sufficiently developed at the time of viability for transmission and perception of painful stimuli [1–6]. Neonates demonstrate physiologic, behavioral, and hormonal responses to painful interventions similar to those seen in older children and adults [7–9]. In addition, neonates exposed to pain early in life react to subsequent noxious stimuli in a way that suggests that they may have some form of memory of their first painful experience [10,11]. Despite evidence, surveys on pain management in neonates among health care professionals indicate either a lack of awareness of pain perception in neonates or lack of an explicit plan of action for alleviation of procedural pain [12,13].

Tracheal intubation is a procedure that is frequently performed in neonates. It involves placing an endotracheal tube into the trachea via the nares or the mouth using laryngoscopy. Most intubations are performed under emergency conditions as part of the initial resuscitation or semiselectively for poor respiratory effort (apnea). The use of premedication or anesthesia is often a secondary consideration in the above situations because intravenous access may not be in place or not easily established. Many neonates are also electively intubated during their stay in the neonatal intensive care unit (NICU) for blocked endotracheal tubes,

* Corresponding author.

E-mail address: vshah@mtsinai.on.ca (V. Shah).

worsening of respiratory distress, accidental extubations, or prior to surgical procedures. Intubation for these indications can be performed in a controlled environment where intravenous access is usually in place for premedication.

Currently, most intubations performed in NICUs continue to be “awake intubation” (ie, without preprocedural analgesia), as reflected in the literature. To date, there have been four surveys conducted in NICUs to determine the extent and type of drugs used prior to intubation; three have been published as journal articles [12,14,15] and one in abstract form [16]. The surveys were conducted in the United States [15], Canada [12,16], and the United Kingdom [14]. Zeigler and Todres [15] reported that in 74 of the 101 university-affiliated NICUs in the United States that responded, 84% never and 97% only rarely used sedatives or muscle relaxants to facilitate intubation. Of the respondents, 82% indicated that the use of atropine prior to intubation was not necessary [15]. Similarly, Fernandez and Rees [12] noted that in Canadian level 3 NICUs, 88% rarely and 84% never used an analgesic for “emergent” or “elective” intubations. The corresponding figures for sedatives were 85% and 73%, respectively [12]. These figures were based on the information provided by the nurses in charge of 30 Canadian level 3 NICUs. A more recent study from the United Kingdom [14] reported that only 37% of the NICUs used any sedation prior to intubation, only 14% of the NICUs had a formal protocol for premedication, and only 8% of the NICUs used drugs for elective or emergent reintubations. In a more recent Canadian survey [16], information was collected based on gestational age: group 1, <30 weeks, group 2, 30 to 37 weeks, and group 3, >37 weeks. Some premedication was used in 68%, 76%, and 87% of the intubation procedures in groups 1, 2, and 3, respectively. Physicians were more likely to use premedication for neonates of higher gestational age, and it was almost exclusively used for elective intubation. The preferred analgesics, anxiolytics, and neuromuscular blockers were fentanyl (66%–68%) and morphine (53%–57%), midazolam (38%–44%), and succinylcholine (45%–51%), respectively. Predominant reasons for not using premedication were “insufficient evidence,” “no need,” and “lack of comfort of using drugs” [16]. These surveys indicate marked variation in the use of premedication for intubation in neonates. To the reviewers’ knowledge, the evidence for the effectiveness and harms associated with premedicated versus awake intubation in neonates has not been the topic of a systematic review.

Physiologic effects associated with premedication for intubation in neonates

In adults and children, awake intubation is known to be associated with pain and alterations in physiologic responses such as hypertension and tachycardia [17–19] and an increase in the common carotid artery and middle cerebral artery blood flow velocity, with a decrease in the femoral blood flow velocity [20].

Similar changes in physiologic responses with intubation have been studied and are well documented in neonates. The laryngoscope blade and the tube can stimulate vagal and mechanical pressure reflexes causing bradycardia, hypoxemia, elevation of systolic blood pressure, and increase in intracranial pressure [21–24]. Concerns have been raised that elevation in the intracranial pressure and systemic hypertension associated with intubation can contribute to intraventricular hemorrhage. Changes in physiologic responses to awake intubation are consistent with the hypothesis that neonates experience pain, and favor the use of premedication for intubation.

The objective of this systematic review is to assess the effectiveness and potential adverse effects of premedication for endotracheal intubation in neonates.

Literature search

MEDLINE was searched for relevant articles published from 1966 to November 2001, Embase from 1980 to November 2001, and CINAHL from 1982 to November 2001 using the following MeSH terms or text words in different combinations: “infant-newborn,” “pain,” “endotracheal intubation,” “premedication,” “neonates,” “analgesia,” “sedation,” “anesthesia,” and “muscle relaxants;” and specific keywords for certain interventions such as “fentanyl,” “morphine,” “thiopental,” “atropine,” “succinylcholine,” and “pancuronium.” The Cochrane Library (Issue 4, 2001) was searched to identify relevant randomized controlled trials. In addition, manual searches of bibliographies and personal files were performed. No language restrictions were applied.

Inclusion criteria

Randomized controlled trials and cohort studies were included. Studies in neonates that provided data either on the number of subjects enrolled or the number of intubation procedures performed were included. Because neonates respond to noxious stimuli with behavioral, physiologic, hormonal, and metabolic alterations, any study that reported data on one of these variables was included. Other reported outcomes such as time to successful intubation and number of intubation attempts were included. Adverse effects such as chest wall rigidity, trauma, need for resuscitation, and any unforeseen complications reported by the authors were also collected. Clinically more relevant outcomes such as duration of ventilation, upper airway obstruction (subglottic stenosis), and long-term (18–24 months’) neurodevelopmental status were also abstracted. Through a consensus process, the reviewers made a decision to include or exclude a specific study.

Data abstraction

Data from each study were abstracted and checked for accuracy by the reviewers.

Table 1

Characteristics of included randomized controlled trials (based on number of subjects) of premedication for neonatal intubation

Reference	Patient population	Sample size	Interventions	Outcomes	Statistical findings between groups
Bhutada et al [26]	Neonates weighing more than 2 kg at birth and requiring semielective intubation for management of respiratory failure or before surgery	N = 30	Study group (n = 15): thiopental (6 mg/kg)	HR, BP, and oxygen saturation continuously monitored and recorded 10 min before, during, and 20 min after the intubation	During intubation, HR increased to a greater degree ($P < 0.03$) and mean BP increased to a lesser degree in infants ($P < 0.002$) who received thiopental
			Control group (n = 15): placebo (physiologic saline)		
			Intubation was performed 1 min after completion of the infusion to coincide with the peak effect of thiopental	Time to intubation was measured from the introduction of nasotracheal tube into the nostril until its fixation with tape	No significant differences in oxygen saturation between the two groups during and after intubation compared with baseline
	All intubations were nasotracheal and 24 of 30 neonates were evaluated during a change from orotracheal to nasotracheal tube			Data on two infants in the control group and one in the study group were lost during acquisition and, therefore, were excluded from the analysis	Time to intubation was shorter for the study group compared with the control group [mean (SE) 2.7 (0.37) versus 5.08 (1.10) min; $P < 0.04$]
Charlton and Greenhough [27]	The study included 45 neonates and 15 infants evaluated during anesthesia for surgery	N = 45	Neonates were randomly assigned to one of three groups:	HR and systolic and diastolic BP were measured before and after intubation	BP did not increase in response to tracheal intubation in neonates intubated awake

Data on the neonatal population are presented in this table			Group A (n = 15): awake intubation (one neonate received intramuscular atropine)		No differences in the BP between the awake and the two anesthetized groups after intubation
			Group H (n = 15): after N ₂ O and halothane inhalation (13 neonates received intramuscular atropine)		HR in the awake group did not increase significantly after intubation
			Group T (n = 15): after thiopentone and either atracurium or pancuronium		HR increased significantly in groups H and T when values were compared before and after intubation
					Significant changes in BP occurred in group H when values were compared before and after intubation
Friesen et al [28]	Preterm neonates (postconception age <37 wk, weight <2500 g) requiring surgical procedures	N = 12	Group 1 (n = 6): atropine (0.02 mg/kg) (awake intubation)	AFP, a noninvasive indicator of ICP, was measured using a Ladd monitor	AFP increased significantly during awake intubation from 7.7 to 23.8 cm H ₂ O ($P < 0.05$) compared with the anesthetized group (5.8 to 7.0 cm H ₂ O)
			Group 2 (n = 6): atropine (0.02 mg/kg), pancuronium (0.1 mg/kg), and one of the four anesthetics [0.75% isoflurane	HR, systolic BP, and mean BP were recorded at 1-min intervals	Systolic BP increased significantly ($P < 0.05$) during awake intubation

(continued on next page)

Table 1 (continued)

Reference	Patient population	Sample size	Interventions	Outcomes	Statistical findings between groups
			(n = 3), 0.5% halothane (n = 1), 20 µg/kg fentanyl (n = 1), or 2 mg/kg ketamine (n = 1) with intubation after 10 min of mask ventilation		
Kelly and Finer [29]	Neonates with birth weights from 580 to 3450 g (25 – 40 wk) during nasotracheal intubation	N = 30	Group 1 (n = 10): control (no medication) Group 2 (n = 10): atropine (0.01 mg/kg) Group 3 (n = 10): atropine (0.01 mg/kg) and pancuronium (0.1 mg/kg)	HR, BP, tcPO ₂ , and ICP were monitored Baseline recording was obtained for a minimum of 5 min prior to the procedure or to any drug administration and then continuously throughout the procedure, followed by a further 5 min of stabilization period	Infants receiving atropine had significant increases in HR from baseline period prior to drug administration compared with HR just prior to intubation ($P < 0.05$), as did infants receiving atropine and pancuronium ($P < 0.01$) A statistically significant decrease in HR was associated with intubation in groups 1 ($P < 0.001$) and 2 ($P < 0.01$) whereas infants in group 3 ($P > 0.05$) did not experience a decrease There was a significant decrease in tcPO ₂ ($P < 0.02$) associated with increases in mean BP ($P < 0.01$) and ICP

					($P < 0.01$) with intubation in ($P < 0.01$) with intubation in all three groups of infants Pancuronium and atropine administration was associated with lesser changes in ICP ($P < 0.05$) and with least changes in HR in response to intubation All infants experienced an increase in mean BP during intubation procedure No significant differences in systemic BP and $tcPO_2$ between groups were noted
Khammash et al [31]	Neonates requiring nonemergent naso- tracheal intubation	N = 28	Group 1 (n = 6): atropine (0.02 mg/kg) Group 2 (n = 8): atropine (0.02 mg/kg)/ fentanyl (0.005 mg/kg) Group 3 (n = 7): atropine (0.02 mg/kg)/ succinylcholine (2 mg/kg)	HR, BP, oxygen saturation, and $tcPO_2$ were monitored before, during, and after intubation	Intubation time was significantly reduced with succinylcholine and/or fentanyl versus atropine alone (22 $\pm$ 7 s, group 2, 25 $\pm$ 10 s, group 4, 27 $\pm$ 7 s, group 3, and 50 $\pm$ 22 s, group 1; $P < 0.05$)

(continued on next page)

Table 1 (continued)

Reference	Patient population	Sample size	Interventions	Outcomes	Statistical findings between groups
			Group 4 (n = 7): atropine (0.02 mg/kg)/ succinylcholine (2 mg/kg)/fentanyl (0.005 mg/kg)	Duration of the procedure of intubation Three patients were excluded—all in the atropine/fentanyl group Reasons for exclusion were (1) time of intubation was not recorded in one infant, (2) malfunction of dynamap machine and BP not recorded for 3 min after intubation in one infant, and (3) oral intubation was performed in one infant because of significant muscular and chest wall rigidity	The mean BP increased by $\geq 20\%$ from baseline after intubation in the atropine and atropine/succinyl-choline groups ($P < 0.05$), by 12% in the atropine/fentanyl group ($P = 0.06$), and by 4% in the atropine/succinyl-choline/fentanyl group ($P = 0.72$) Significant chest wall rigidity was observed in three patients receiving atropine/fentanyl
Pokela and Koivisto [30]	Newborn infants requiring elective intubation during first month of life	N = 20	Group 1 (n = 10): pethidine (1 mg/kg) administered 15 min prior to intubation	HR, RR, tcPO_2 and/or oxygen saturation, mean BP, and general activity	Hypoxemia during intubation was found in all 10 infants in group 1 and in 7 of 10 infants

Group 2 (n = 10): alfentanil (20 µg/kg) administered just prior to intubation and suxamethonium (1.5 mg/kg)	was monitored continuously from 10 min before intubation to 1 h after intubation	in group 2. The duration of hypoxemia was significantly longer in group 1 (4 min versus 1.5 min; $P = 0.036$)
All infants received glycopyrrolate (3–5 µg/kg)	Recordings were made at 1-min intervals before and during intubation and 5-min intervals thereafter	Intubation was successful on first attempt in 3 of 10 infants in group 1 compared with 7 of 10 infants in group 2
	Serum cortisol, β-endorphin levels, blood glucose, and blood gas analyses were performed prior to intubation and 1 and/or 2 h after intubation	Median duration of intubation was twice as long in group 1 compared with group 2 (120 s versus 60 s) ($P = 0.012$)
		There were no statistically significant differences in the plasma β-endorphin and cortisol values between or within groups

Abbreviations: AFP, anterior fontanel pressure; BP, blood pressure; HR, heart rate; ICP, intracranial pressure; N₂O, nitric oxide; RR, respiratory rate; tcPO₂, transcutaneous oxygen tension.

Statistical methods

If two or more studies assessed the same intervention and reported on the same outcome data, the reviewers planned to combine the data using the RevMan 4.1 (Update Software, Oxford, UK).

Results

When the retrieved literature was critically appraised, it was identified that several studies included infants older than 28 days. In view of the paucity on the review topic, a decision was made to include these studies because they provided relevant information.

Randomized controlled trials

Nine randomized controlled trials were identified, seven as full publications [22,25–30], one in abstract form [31], and one as a letter to the editor [32]. Seven of these trials [26–32] were eligible for inclusion in this review, whereas two were excluded [22,25]. One study was excluded because five additional non-randomized subjects received study medication and were included in the analysis [25]. In the second excluded study, a discrepancy was noted in the study age group between the abstract and the methods section of the article [22].

Detailed information on included and excluded trials is provided in Tables 1–3.

Among the seven included trials, Bhutata et al [26] evaluated the effectiveness of thiopental on heart rate, blood pressure, and oxygen saturation during intubation in 30 neonates. Data on three infants (two in the control group and one infant in the study group) were lost during acquisition and were excluded from the final analyses. Heart rate and blood pressure values in infants premedicated with thiopental were maintained nearer to baseline values. Time taken to intubation was significantly shorter compared with infants who received no premedication.

In the study by Charlton and Greenhough [27], term neonates were randomized either to awake intubation, nitric oxide and halothane inhalation, or thiopentone and a muscle relaxant (atracurium or pancuronium). There was no elevation in blood pressure in infants intubated awake. No differences in the blood pressure between awake and the two anaesthetized groups after intubation were noted.

Friesen et al [28] evaluated the anterior fontanel pressure and other physiologic parameters including heart rate and blood pressure in 12 preterm neonates randomized to receive atropine alone or atropine and pancuronium and one of four anesthetic agents (0.75% isoflurane, 0.5% halothane, 20 µg/kg fentanyl, or 2 mg/kg ketamine). Both anterior fontanel pressure and systolic blood pressure were elevated in infants intubated awake.

Similarly, Kelly and Finer [29] randomized 30 neonates to awake intubation, atropine, or a combination of atropine and pancuronium. Infants who received atropine and pancuronium demonstrated lesser changes in intracranial pressure

Table 2
Characteristics of included randomized controlled trial of premedication for neonatal intubation (based on number of intubation procedures)

Reference	Patient population	Sample size	Interventions	Outcomes	Statistical findings between groups	Adverse effects
Attardi et al [32]	Preterm infants	N = Eight infants underwent 16 intubation procedures	Group 1 (n = 3 intubations): placebo	Physiologic stability (HR, BP, and oxygen saturation recorded at 10-min intervals)	Number of episodes of oxygen desaturation around the time of intubation was significantly greater in group 3 (86%) than in group 1 (0%); $P = 0.01$	Study terminated due to concerns regarding adverse events
			Group 2 (n = 6 intubations): atropine and placebo	Success rate of intubation		Cardiopulmonary resuscitation was required in 29% of group 3 compared with 0% in group 2 ($P = 0.16$)
			Group 3 (n = 7 intubations): atropine and midazolam			

Abbreviations: BP, blood pressure; HR, heart rate.

Table 3
Characteristics of excluded trials of premedication for neonatal intubation

Reference	Patient population	Sample size	Interventions	Outcomes	Statistical findings between groups	Reasons for exclusion
Barrington et al [25]	Preterm newborn infants requiring intubation for clinical indications and who had an indwelling arterial catheter in place	N = 20, randomized using computer-generated code	Group 1 (n = 10): atropine (20 µg/kg)	HR, MAP, tcPO ₂ , and ICP monitored continuously before, during, and after intubation	No infants in either group suffered bradycardia (HR < 100 beats/min) during intubation	Not a true randomized controlled study
		Further five nonrandomized infants evaluated because randomization procedure failed to produce well-matched group	Group 2 (n = 10): succinylcholine (2 mg/kg) plus atropine		The lowest tcPO ₂ in group 1 was significantly lower than the lowest tcPO ₂ in group 2 ($P < 0.05$)	Data on randomized and nonrandomized infants who received group 1 protocol were combined
					Increase in ICP developed during intubation in infants receiving atropine alone but was prevented by premedication with succinylcholine and atropine ($P < 0.01$)	
					A 41% increase in systemic BP occurred immediately after administration of succinylcholine ($P < 0.01$)	

				Intubation was significantly shorter in infants receiving succinylcholine ($P < 0.05$)	
Millar and Bissonnette [22]	Neonates aged between 1 and 34 d	N = 13	Group 1 (n = 7): awake intubation Group 2 (n = 6): thiopentone (5 mg/kg) and succinylcholine (2 mg/kg) Following baseline observations, all patients received preoxygenation for 3 min and atropine (0.02 mg/kg)	HR, systolic BP, AFP, CBFV, and oxygen saturation were recorded at following intervals: baseline, after intravenous atropine 0.02 mg/kg, during laryngoscopy, immediately after insertion of the ETT, and 1 and 5 min later At laryngoscopy, change in the HR was higher in group 1 compared with group 2 ($P < 0.05$) There were no differences between groups with respect to changes in the systolic BP; however, both groups showed increases in the BP at the time of laryngoscopy ($P < 0.05$) There was an increase in AFP from resting values at the time of laryngoscopy in group 1 ($P < 0.05$) No differences in the values of CBFV between the groups or within the groups	Discrepancy in the study age group in the abstract (aged 1–34 d) and in the methods section (aged 1–42 d) of the article

Abbreviations: AFP, anterior fontanel pressure; BP, blood pressure; CBFV, cerebral blood flow velocity; ETT, endotracheal tube; HR, heart rate; ICP, intracranial pressure; MAP, mean arterial pressure; tcPO₂, transcutaneous oxygen tension.

Table 4

Characteristics of included cohort study of premedication for neonatal intubation (based on number of subjects)

Reference	Patient population	Interventions	Outcomes	Results
Naulaers et al [34]	18 newborn infants with severe respiratory or cardiac conditions Mean gestational age, 36.3 wk (range 32–42 wk); mean postnatal age, 23 d (range 1–150 days); and mean weight, 2588 g (range 1390–5000 g) at the time of the study	Short-acting barbiturate methohexital in a dose of 2.6 mg/kg	Speed of action and the degree of relaxation, sedation, and sleep in the first 5 min after administration	Sedation was achieved in all patients during the first 2 min, 11 of 18 infants remained well sedated after 3 min, and 4 of 18 infants did not show any reaction after 5 min All patients moved spontaneously after 10 min
			Observations were carried out every minute for the first 5 min	
			The degree of sedation was assessed by motor response to external stimuli (rubbing the heel with the swaddle)	Relaxation was achieved during the first 2 min in all patients, 12 of 18 infants remained hypotonic after 3 min, and 1 infant became hypertonic after 5 min, remaining hypertonic for 1 min
			Relaxation was evaluated by clinical evaluation of the tone in arms and legs HR, systolic and diastolic BP, and oxygen saturation were recorded every 30 s for the first 2 min, every minute up to 5 min, and every 5 min thereafter for the next half hour	All infants had normal tone after 8 min and all were awake after 12 min No short-term or long-term effects were noted on the HR
			Observations were made for hiccups, abnormal behavior, muscle twitching, convulsions, and other side effects	Except for a transient drop in systolic BP between 1 and 3 min, no important changes were noted in the BP The most important side effect seen was muscle twitching in two patients, which never lasted longer than 1 min nor did it reoccur Three patients developed hiccups

Abbreviations: BP, blood pressure; HR, heart rate.

and the least changes in heart rate in response to intubation. There were no significant differences in transcutaneous oxygen tension and systemic blood pressure between groups.

Khammash et al [31] randomized 28 neonates to one of four groups: atropine, atropine and succinylcholine, atropine and fentanyl, or all three drugs. The time taken to intubate the neonates was significantly shorter in the group that received succinylcholine and/or fentanyl compared with atropine alone. The mean blood pressure increased the least in the group that received all three drugs. Clinically significant chest wall rigidity occurred in three infants who received atropine and fentanyl in combination.

Pokela and Koivisto [30] evaluated physiologic and hormonal responses to intubation in 20 neonates who were randomized to receive either pethidine or a combination of suxamethonium and alfentanil. Hypoxemia was noted in all 10 neonates who received pethidine compared with 7 of 10 neonates who received alfentanil and suxamethonium ($P=0.036$). There were no differences in the plasma β -endorphin and cortisol values between or within groups before intubation or 1 and/or 2 hours after intubation.

Attardi et al [32] randomized 8 infants undergoing 16 intubation procedures. They were randomized into 3 groups: placebo, atropine and placebo, and atropine and midazolam. The administered doses of the drugs were not reported. The study was terminated early because of concerns regarding adverse reactions. Cardiopulmonary resuscitation was required in 29% of the intubations in the atropine and midazolam group compared with 0% in the atropine/placebo group.

Cohort studies

Two studies were identified and included in this review [33,34]. Details of these studies are presented in Tables 4 and 5.

Naulaers et al [34] evaluated the effectiveness of a short-acting barbiturate methohexital for its speed of action and degree of relaxation, sedation, and sleep in the first 5 minutes of administration in 18 neonates undergoing intubation. Sedation and relaxation were achieved in all neonates in the first 2 minutes. No short-term or long-term effects were seen on heart rate or blood pressure. Important side effects that were observed were muscle twitching and hiccups.

Barrington and Byrne [33] administered atropine, fentanyl, and succinylcholine for 253 consecutive intubation procedures during an 18-month study period. Of the 253 intubations, 194 were uneventful, 28 required two attempts, 22 required more than two attempts and, in 9, a second attempt with a smaller-sized endotracheal tube was required. Four infants developed chest wall rigidity after administration of fentanyl. In three infants, the chest wall rigidity resolved with the use of succinylcholine, whereas in one infant, the chest wall rigidity continued despite the use of succinylcholine and required immediate intubation.

In summary, anticholinergics, analgesics, anaesthetic agents, muscle relaxants, sedatives, and amnesic agents were used in isolation or in various combinations for premedication in neonates. The most commonly evaluated combination was that of atropine, fentanyl, and succinylcholine. The majority of these studies were

Table 5

Characteristic of included cohort study of premedication for neonatal intubation (based on number of intubation procedures)

Reference	Patient population	Interventions	Outcomes	Results
Barrington and Byrne 1998 [33]	269 consecutive nasotracheal intubations occurred during the study period (18 months)	Premedication consisted of atropine 20 µg/kg intravenously, followed by fentanyl 3–4 µg/kg and immediately prior to intubation succinylcholine in the dose of 2 mg/kg	Success rate, incidence of complications and adverse effects of a policy of routine premedication for endotracheal intubation	253 intubations were premedicated and on 16 occasions premedication was not given 178 intubations were performed in the NICU, 11 of which were not given premedication 75 intubations were performed during transport, of which 5 did not receive premedication 5 infants had major airway abnormalities and all of these received premedication and intubations were performed by staff neonatologists 5 infants who had intubations while they were on muscle relaxants and receiving narcotic infusions were excluded
	Premedication was administered if an IV was in place and if the intubation was not an absolute emergency (ie, the patient could not be stabilized by bag and mask IPPV)			
	Gestational age of these infants ranged from 22–42 weeks gestation			
	Postnatal age ranged from 30 minutes to 192 days			

Of the 253 intubations using premedication, 194 were without incident, 28 required two attempts, 22 required more than 2 attempts and 9 required a second attempt with a smaller endotracheal tube

On 4 occasions infants developed chest wall stiffness resulting in difficulty with bag and mask ventilation and associated low oxygen saturations

In 3 of the 4 occasions rigidity resolved with administration of succinylcholine

On one occasion, infant remained desaturated despite succinylcholine; intubation was performed immediately, and the oxygenation improved

Abbreviations: NICU = Neonatal Intensive Care Unit.

conducted under relatively controlled circumstances such as elective or semi-elective intubation (surgery or change of endotracheal tube). In one study, premedication was used during transport [33]. Side effects such as chest wall rigidity were associated with fentanyl [31,33], and an increased need for cardiopulmonary resuscitation occurred with midazolam [32]. The dose of midazolam used for premedication was not reported in the study by Attardi et al [32].

Because the authors of the published studies used different drugs and outcome measures, the reviewers were unable to identify two studies from which the data could be summarized using meta-analytic techniques.

Discussion

There is currently limited evidence available regarding the efficacy/effectiveness of different drugs used for premedication in neonates. In general, the studies that were reviewed enrolled a small number of neonates. Results of some of the reviewed trials suggest that there is attenuation of the physiologic responses in neonates who received premedication for intubation; it also shortens the time for the procedure and makes intubation easier. None of the studies, however, address more clinically relevant short-term and/or long-term outcomes such as the duration of ventilation, occurrence of subglottic stenosis, or neurodevelopment. The long-term effects of repeated exposure to premedication for frequent intubations in the same neonate has not been studied.

There is substantial evidence that neonates are capable of perceiving pain and that pain during early life may influence pain thresholds and responses during later life. Neonates have no ability to verbally express pain; therefore, physiologic and behavioral measurements are used as surrogate markers for pain. Because physiologic responses are positively influenced by use of premedication, it has been inferred that its use during intubation is beneficial. None of the studies reviewed used behavioral indicators to assess pain. Given that behavioral or composite measures are considered the optimal approach to assessing pain in neonates, it could be inferred that pain was not adequately evaluated in these studies. Whether decreasing the immediate physiologic and behavioral responses to intubation translates into long-term benefits or harms is unknown.

Individual NICUs need to develop specific practice guidelines regarding premedication for intubation to ensure that all staff are familiar with the effects of the drugs being used. Ongoing monitoring when these drugs are used should be in place. Under certain circumstances where the risks associated with premedication outweigh the potential benefits, such as in a neonate with suspected abnormal upper airway anatomy (eg, severe micrognathia or previously diagnosed subglottic stenosis), awake intubation may be acceptable. If a decision is made to administer premedication, then it should be done with caution in a setting where the infant's cardiorespiratory status is being monitored and in the presence of skilled individuals.

Future research should be targeted to document the safety and effectiveness of drugs for premedication in both stable and unstable preterm and term neonates and

for intubations performed under emergent and controlled situations. Long-term neurologic outcomes of these neonates need to be included in the study design.

Summary

Implications for practice

Extrapolating information from the adult and pediatric literature suggests that awake intubation is probably inappropriate in most neonates. Because premedication attenuates the physiologic responses to intubation, its use is recommended. Adequately skilled staff who have a full understanding of the potential benefits and harms of the interventions used should perform intubation and the administration of premedication in neonates.

Implications for research

There is a need for well-designed and well-executed randomized controlled trials assessing the effectiveness and potential adverse effects of premedicated intubation in neonates. A valid pain assessment measure or approach should be used. Both short-term and long-term physiologic and clinical outcomes should be incorporated into the trial design.

References

- [1] Anand KJ, Carr DB. The neuroanatomy, neurophysiology and neurochemistry of pain, stress and analgesia in newborns and children. *Paediatr Clin N Am* 1989;36(4):795–822.
- [2] Fetus and Newborn Committee. Canadian Paediatric Society, and Committee on Fetus and Newborn, Committee of Drugs, Section of Anaesthesiology and Section on Surgery, American Academy of Paediatrics: prevention and management of pain and stress in the neonate. *Paediatr Child Health* 2000;5(1):31–8.
- [3] Fitzgerald M, Shaw A, McIntosh N. Postnatal development of the cutaneous flexor reflex: comparative study of preterm infants and newborn rat pups. *Dev Med Child Neurol* 1988; 30(4):520–6.
- [4] Fitzgerald M, Millard C, McIntosh N. Cutaneous hypersensitivity following peripheral tissue damage in newborn infants and its reversal with topical anaesthesia. *Pain* 1989;39(1):31–6.
- [5] Fitzgerald M, Anand KJS. Developmental neuroanatomy and neurophysiology of pain. In: Schechter NL, Berde CM, Yaster M, editors. *Pain in infants, children and adolescents*. Baltimore (MD): Williams & Wilkins; 1993. p. 11–32.
- [6] Giannakouloupoulos X, Sepulveda W, Kourtis P, et al. Fetal plasma cortisol and beta-endorphin response to intrauterine needling. *Lancet* 1994;344(8915):77–81.
- [7] Anand KJ, Hickey PR. Pain and its effects in the human neonate and fetus. *N Engl J Med* 1987; 317(21):1321–9.
- [8] Anand KJ, Sippell WG, Anysley-Green A. Randomised trial of fentanyl anaesthesia in preterm babies undergoing surgery: effects on the stress response. *Lancet* 1987;I(8524):62–6.
- [9] Anand KJ, Hickey PR. Halothane-morphine compared with high-dose sufentanil for anesthesia and postoperative analgesia in neonatal cardiac surgery. *N Engl J Med* 1992;326(1):1–9.
- [10] Taddio A, Goldbach M, Ipp M, et al. Effect of neonatal circumcision on pain responses during vaccination in boys. *Lancet* 1995;345:291–2.

- [11] Taddio A, Katz J, Ilersich AL, et al. Effect of neonatal circumcision on pain response during subsequent routine vaccination. *Lancet* 1997;349:599–603.
- [12] Fernandez CV, Rees EP. Pain management in Canadian level 3 neonatal intensive care units. *Can Med Assoc J* 1994;150(4):499–504.
- [13] Franck L, Lund C, Fanaroff A. A neonatal survey of the assessment and treatment of pain in the newborn intensive care unit. *Paediatr Res* 1986;20:347A.
- [14] Whyte S, Birrell G, Wyllie J. Premedication before intubation in UK neonatal units. *Arch Dis Child (Fetal Neonatal Ed)* 2000;82:F38–41.
- [15] Ziegler JW, Todres D. Intubation of newborns. *Am J Dis Child* 1992;146(2):147–9.
- [16] Vogel S, Gibbins S, Simmons B, et al. Premedication for endotracheal intubation (EI) in neonates: a Canadian perspective. *Paediatr Res* 2000;47(4):438A.
- [17] King BD, Harris LC, Greifenstein FE, et al. Reflex circulatory responses to direct laryngoscopy and tracheal intubation performed during general anesthesia. *Anesthesiology* 1951;12:556–66.
- [18] Shribman AJ, Smith G, Achola KJ. Cardiovascular and catecholamine responses to laryngoscopy with and without tracheal intubation. *Br J Anaesth* 1987;59:295–9.
- [19] Stoelting RK. Circulatory changes during direct laryngoscopy and tracheal intubation: influence of duration of laryngoscopy with or without prior lidocaine. *Anesthesiology* 1977;47:381.
- [20] Moorthy SS, Greenspan CD, Dierdorf SF, et al. Cerebral blood flow increases and femoral blood flow decreases during laryngoscopy and tracheal intubation. *Anesthesiology* 1993;77(3A):A73.
- [21] Marshall TA, Deeder R, Pai S, et al. Physiological changes associated with endotracheal intubation in preterm infants. *Crit Care Med* 1984;12(6):501–3.
- [22] Millar C, Bissonnette B. Awake intubation increases intracranial pressure without affecting cerebral blood flow velocity in infants. *Can J Anaesth* 1994;41(4):281–7.
- [23] Raju TNK, Vidyasagar D, Torres C, et al. Intracranial pressure during intubation and anesthesia in infants. *J Paediatr* 1980;96(5):860–2.
- [24] Stow PJ, McLeod ME, Burrows FA, et al. Anterior fontanelle pressure responses to tracheal intubation in the awake and anesthetized infant. *Br J Anaesth* 1988;60:167–70.
- [25] Barrington KJ, Finer NN, Etches PC. Succinylcholine and atropine for premedication of the newborn infant before nasotracheal intubation: a randomized controlled trial. *Crit Care Med* 1989;17(12):1293–6.
- [26] Bhutata A, Sahni R, Rastogi S, et al. Randomised controlled trial of thiopental for intubation in neonates. *Arch Dis Child (Fetal Neonatal Ed)* 2000;82(1):F34–7.
- [27] Charlton AJ, Greenhough SG. Blood pressure response of neonates to tracheal intubation. *Anaesthesia* 1988;43(9):744–6.
- [28] Friesen RH, Honda AT, Thieme RE. Changes in anterior fontanel pressure in preterm neonates during tracheal intubation. *Anesth Analg* 1987;66(9):874–8.
- [29] Kelly MA, Finer NN. Nasotracheal intubation in the neonate: physiologic responses and effects of atropine and pancuronium. *J Paediatr* 1984;105(2):303–9.
- [30] Pokela ML, Koivisto M. Physiological changes, plasma β -endorphin and cortisol responses to tracheal intubation in neonates. *Acta Paediatr* 1994;83:151–6.
- [31] Khammash HM, O'Brien K, Dunn MS, et al. Blunting of hypertensive response to endotracheal intubation in neonates by premedication. *Paediatr Res* 1993;33(4):218A.
- [32] Attardi DM, Paul DA, Tuttle DJ, et al. Premedication for intubation in neonates. *Arch Dis Child (Fetal Neonatal Ed)* 2000;83(2):F161.
- [33] Barrington KJ, Byrne PJ. Premedication for neonatal intubation. *Am J Perinatol* 1998;15(4):213–6.
- [34] Naulaers G, Deloof E, Vanhole C, et al. Use of methohexital for elective intubation in neonates. *Arch Dis Child* 1997;77:F61–4.